CREATIVE RESPONSES TO SOUNDSCAPE ECOLOGY:

INNOVATIVE FRAMEWORKS AND CASE STUDY

Jesse Budel

Elder Conservatorium of Music
Faculty of Arts, The University of Adelaide

ABSTRACT

Soundscape ecology is becoming a pioneering discipline in contemporary ecological research, investigating the relationships between landscape and soundscape (or topographic and acoustic patterns). In recent years, the discipline has enjoyed increasing international attention as a prominent field of research in contemporary ecology, with a proliferation of associated publications and conferences.

The breadth and scope of this field offers many possibilities for innovative creative applications. Whilst there is already a strong body of artistic work with the humanities-oriented acoustic ecology movement and other ecoacoustic compositional approaches, the comprehensive frameworks and methodologies of soundscape ecology have thus far not been specifically explored in a creative context. This paper explores a creative process-in-development that adapts these principles of soundscape ecology, and presents a compositional case study in response to Mobilong Swamp (near Murray Bridge, South Australia), and its related soundscape and ecosystem.

1. INTRODUCTION

Soundscape ecology is becoming a pioneering discipline in contemporary ecological research, in recent years enjoying increasing international attention through significant interdisciplinary publications and collaborations, and the formation of related societies.

The breadth and scope of this field offers many possibilities for innovative creative applications. Through the rise of environmental activism and technological development in recent decades, there is already a strong body of artistic work with the humanities-oriented acoustic ecology movement (notably, through the soundscape composition work of Truax, Westerkamp and others) and multiple other ecoacoustic compositional approaches. Even so, the relative novelty of soundscape ecology has meant that comprehensive principles, frameworks and methodologies of soundscape ecology have thus far not been specifically explored in a creative context - a gap that this paper endeavours to address.

First, an outline of the foundational principles of soundscape ecology is considered, establishing its interdisciplinary background, analytical concerns and methodologies. A creative framework-in-development adapting the discipline is subsequently explored, with attention given to each stage of the process. A case study in response to Mobilong Swamp, near Murray Bridge, South Australia is subsequently presented, with discussion around its environmental history and current ecological status, and application of the creative process to the site.

1.1. Background

Soundscape ecology is the study of the relationship between landscape and soundscape, or topographic and acoustic patterns. It "[describes] complex phenomena at community, ecosystem and landscape scales within natural and human dominated systems" (Farina 2014).

First proposed in a 2011 *Bioscience* article (Pijanowski et al.), and followed soon after with a special issue of *Landscape Ecology* (Pijanowski and Farina 2011), soundscape ecology garnered attention as an innovative and promising field in contemporary ecology. In recent years, a seminal textbook (Farina 2014) and establishment of the related International Society of Ecoacoustics has done much to further broader interest and research.

Drawing on research in the areas of spatial (or landscape) ecology, psychoacoustics, bioacoustics, and acoustic ecology, the discipline takes an interdisciplinary approach, representing perspectives from ecology, psychology, ethology and the humanities respectively. It should be noted that there is inclusion of a cognitive element here, as "the description of soundscape patterns is an indispensable, but not sufficient, way of studying (ecologically) this matter; a biosemiotic approach is also required to understand and interpret the uses and functions of sounds" (Farina and Pieretti 2012, 22). As such, the consideration of both active contributors and passive observer in soundscape participation and interpretation is integral.

1.2. Analysis

Through drawing on the multitude of knowledges from each of the above disciplines, soundscape ecology is able to form a coherent and comprehensive understanding of a soundscape, its component parts, and underlying ecological processes and relationships.

There are three categories by which sounds may be basically described (Krause 2002):

biophony - is the collection of sounds produced by all organisms at a location over a specified time
geophony - are the sounds originating from the geophysical environment, which includes wind, water, thunder, movement of earth, etc.
anthropophony (also anthrophony/technophony) - human produced sounds, created by stationary and moving human-made objects

Additionally, there are a number of parameters that are considered in the analysis of soundscapes, including:

1.2.1. Frequency Patterns

As organisms vocalise in broad sonic spectra, their individual and collective communication systems have several suspected ecological causes. Inspecting the behaviour of biophonies, soundscape ecology draws on various bioacoustics hypotheses, seeking to better understand and explain how interspecific vocalisations have arisen with respect to evolutionary processes, and related performance and communication strategies in soundscapes.

These include the Morphological Adaptation Hypothesis (correlation between evolution of physiological traits and vocalisation characteristics), Acoustic Adaptation Hypothesis (correlation of vocalisation with communication strategies appropriate to physical environment attributes), Acoustic Niche Hypothesis (explanation of biophonic organisation into frequency and/or temporal niches for communicational efficacy) and the Species Recognition Hypothesis (explanation of different vocalisations between sympatric species to prevent hybridisation).

1.2.2. Temporal Patterns

Many recognised temporal cycles are exhibited in soundscape activity, often reflecting the timing of organisms' life histories. At the relative microlevel, dawn and duck choruses of birds, amphibians, insects and other organisms correlate with diurnal and nocturnal cycles, often motivated by feeding or mating activity. Mesolevel seasonal cycles impact many organisms' activities each year, including breeding, migration and hibernation. Most broadly at the macrolevel, climate patterns and geological activity impacts and alter ecosystem composition and the resultant soundscape.

1.2.3. Spatial dimensions and gradients

In seeking to establish identifiable spatial dimensions in soundscape analysis, the concepts of ecotopes (landscape patches with coincidental geotopes and biotopes) and ecotones (the boundaries between ecotypes) from landscape ecology can be adapted to sonotopes, soundtopes and sonotones:

sonotope - a distinct sonic unit or sonic patch produced by the overlap of geophonies, biophonies and anthropophonies, resulting from each sound's different spatial and temporal extension.
soundtope - a subdivision of a sonotope specifically related to biophony, that inspects the active and passive exchange of information among collaborative members of a community.
sonotone - the boundary between adjacent sonotopes.

Related to a soundscape's spatial dimensions are gradients, which are the gradual changes in abiotic factors in an ecosystem. Through their dynamism, they impact multiple elements of an ecosystem and its soundscape, and include:

altitude - elevation, air pressure, wind, precipitation
flow - rain and stream flows
habitat interior-edge - species distribution, changed weather dynamics (wind speed, turbulence, vorticity)
latitude - annual climate patterns (temperature, solar radiation), climate change
human disturbance - land use and transformation

1.2.4. Interactions

This area of analysis looks at the interactions between biophony, geophony and anthropophony, and the way each responds to the other through agency (by adapting call amplitude, frequency, or timing in animals, or modifying activities in humans) or abiotic process.

Each of these areas of analysis produce specific information that collectively provides a comprehensive understanding of a soundscape's component parts and ecological processes and relationships.

1.3. Methodologies

Soundscape ecology is founded on the methodologies of landscape ecology, which in turn directly employ or appropriate field techniques and methods from ecology and naturalist studies, in addition to those of its other associated disciplines. As Farina writes, the discipline uses a "broad range of field and laboratory approaches from bioacoustics to psychoacoustics and truly heterogenous methods to collect, process, and interpret sonic data' (2014, 221).

As such, multiple tools are employed to develop a comprehensive understanding of a landscape and its acoustic environment. Spatial analysis tools (including geographic information systems (GIS), remote sensing technology and spatial metric software) have allowed landscape ecology to flourish as a discipline, and are accordingly used in soundscape ecology research. Additionally, particular to the study of sound, there are a number of tools that soundscape ecology employs:

1.3.1. Digital sound recording

The efficiency and fidelity of contemporary recording equipment permits high quality sound recordings of a soundscape, resulting in rich data for analysis and interpretation. Microphone type, quantity, direction and placement, and recording system, are selected appropriate to the soundscape type and conditions, and research objectives. Automated recording technologies also permit the recording of soundscapes at specified times and durations, allowing for strategic capture of acoustic data. There are a number of challenges that remain, including storage of large data quantities, equipment maintenance on recording networks, and until recently, equipment cost, which is now

increasingly diminishing (Krause 2002; Monacchi 2011).

1.3.2. Analytical tools, measurement, and metrics

Sound can be measured in a number of ways, and in soundscape ecology, the main goal is "to process data to extract the emerging patterns in terms of complexity/ information of the sonic environment" (Farina 2014, 239). The spectrogram is most commonly used in this regard (mapping frequency, time and amplitude), and other emerging tools related to other parameters (i.e. geographic and topological mapping, sound source identification) are increasingly used.

Additionally, metrics have been developed for evaluation and comparison of different recordings and soundscapes, such as noise and exposure levels, and entropy, richness, dissimilarity, complexity and evenness indices (ibid., 239-47).

1.3.3. Software and databases

The development in audio technology over the past few decades has seen many advances and improvements in the levels of information captured and analysed in sound. Complementing the rise of digital sound recording was the release of increasingly sophisticated software and database packages, which permitted precise analysis, manipulation and rendering of sound samples, correlation of these with spatial interfaces and databases, and calculation of metrics. In soundscape ecology, this has furthered researchers' capacity for ecological enquiry and interpretation. Notable examples include Wave surfer (Sjölander and Beskow 2006), SEEWAVE (Sueur, Aubin, and Simonis 2008), SPEAR (Klingbeil 2009), Raven (Cornell University, 2014), and Pumilio (Villanueva-Rivera and Pijanowski 2012).

2. CREATIVE FRAMEWORK

With the establishment and development of this field comes many opportunities for creative application. Whilst there is already a strong tradition of artistic work with the acoustic ecology movement, and more recently and relevantly, David Monacchi's utilisation of the acoustic niche hypothesis (Krause 1993; Monacchi 2013), the comprehensive principles, frameworks and methodologies of soundscape ecology have thus far not been specifically explored in a creative context.

To address this gap, these principles have been adapted into a creative compositional process, with the intention of engaging meaningfully with specific places and their ecosystems and soundscapes, so as to produce representative and reflective creative responses.

The stages of this creative process are conception, collation, preparation, composition, realisation and reflection.

2.1. Conception

After first selecting the site to engage with, one may begin developing the conceptual foundations of the creative work.

2.1.1. Research

In order to make informed and effective plans throughout the compositional process, it is necessary to research multiple perspectives of the place, by reviewing multiple sources of information (i.e. documentation, maps, field recordings, artefacts, oral transmission, etc). Related to soundscape ecology, this understandably refers to ecological and environmental information; additionally, historical, sociocultural and other sources of information should also be considered, to add context to the site's contemporary environmental situation and associated ecosystems, soundscapes and identity.

2.1.2. Sketching

Following the research, preliminary sketching may take place, which brings together the multiple perspectives of place investigated to provide creative context in the planning stage. This should consider the parameters of the work, including the duration, media (whether acoustic, electroacoustic, or a combination) and performance space, amongst.

At this early stage, engagement with place and investigation of resources should be permitted influence over the final outcome of the work, as per Westerkamp's reflections on the dynamic work cycle of soundscape composition (2002, 54). As such, such sketches need not provide definitive compositional structures, but instead act as preliminary guides as to how the space might be engaged, and with which perspectives it might be examined.

2.1.3. Planning

The research and sketching will inform the plans for the engagement with place and ensuing creative work.

As the soundscape ecologist operates with similar field work approaches as a regular ecologist, considerations regards equipment and logistics will need to be made (appropriate recording, monitoring and documenting equipment; field trip resources including clothing, food, accommodation and amenities) respective to the location's conditions and emerging creative concepts.

Additionally, planning should also be made regarding the eventual presentation of the creative work, involving liaising with relevant ensembles and organisations to successfully present the work.

2.2. Collation

The preparations made during the conception stage can now be applied during the collation stage. Practically, this may also be understood as the engagement with place, where one travels to the location being researched and conducts field work, collecting resources (materials, information and ideas) for the composition. Activities may include:

2.2.1. Sound walks

A predominant activity of acoustic ecology, sound walking on location allows the composer to become familiar with the physical and sonic qualities of the place and soundscape, informed and contextualised by

the information gathered during research in the conception stage (Schafer 1993; Allen 2013).

2.2.2. Observation and documentation

Surveying the place draws on the methodologies of both soundscape ecology and acoustic ecology. The former combines traditional approaches as employed by naturalists and ecologists (documenting geographic, biological and other environmental factors) as well as the contemporary considerations of soundscape ecology (observation of bio-, geo- and anthrophony; examination of bioacoustics hypotheses and effects of gradients, use of available technologies) (Krause 2002; Farina 2014; Krause 2015). The latter, by contrast, encourages more subjective and aesthetic observations (as per the concerns of acoustic ecology (Schafer 1993)).

2.2.3. Field recordings

Capturing an acoustic environment through recording is integral to soundscape ecology activity, and remains the most effective and efficient way of documenting the sonic realities of a place. Many decisions factor into the recording process (type of equipment, microphone position, time of day, duration of take, encoding), which are considered with respect to the intending compositional outcomes, particularly spatialisation (Monacchi 2011).

2.2.4. Consultation/Literature Research

Liaising with people associated with the place of research (living, working, etc.) and reviewing documented materials often reveals more specific and intimate knowledge and perspectives (Schafer 1993).

2.3. Preparation

Following the collation stage, the materials are analysed and transformed in preparation for the composition stage.

2.3.1. Analysis

Initially, analysis of the materials is necessary, so as to not only extract information about the content and context of each resource, but also inform the transformative techniques that will adapt them to a musical context. This might be quantitative analysis, assisted by technology, to measure materials and calculate metrics; or qualitative analysis, involving subjective evaluation of the materials.

Additionally, there is a need in the creative process to relate soundscape ecology analysis to musical concerns, in the interest of preserving content identity (mimetic relationship between materials and source) and context (ecological relationships and processes) (Monacchi 2011; O'Callaghan 2012).

Considering the frameworks of soundscape ecology, it is noted that the discipline analyses a variety of different parameters' patterns, including frequency, temporal, amplitudinal, spatial, and interactional patterns. Music similarly is comprised of multiple elements, defined by Burton as sound, structure and artistic intent (2015, 25). Sound may be qualified into further sub-elements:

pitch - **'**represents how the mind perceives the cycles, repetitive nature of sonic vibrations', i.e. the perception and cognitive organisation of sonic frequencies.
duration - 'represents the onset and offset signals created by nerve responses to sounds', or rather, the perception of temporal phenomena of sound, including pulse, rhythm, and tempo.
loudness - 'represents the totalled number of auditory nerve stimulations over short cyclic time periods'. Perceptively, this relates to amplitudinal phenomena including dynamics and articulation.
timbre - 'represents the product of information gained from frequency transients, noisiness, unsteadiness, perceived pitch and the spread and intensity of overtones in a sound, all of which is used to identify the sound.'

Moving beyond single sound sources and considering the total sonic environment, further sub-elements include:

texture - **'**relates to the number of sound sources and the interaction between them.'
spatial location - 'represents the cognitive placement of a sounds in an environmental context; including the placement of a sound on both the horizontal and vertical place, the distance to the sound source and the characteristics of the sonic environment.'

Complementing sound are two additional musical elements:

structure - an inherent part of the compositional act is synthesising sonic elements into meaningful structures, as, 'if music is to communicate information of any form, sounds need to be structure in an understandable way.' (ibid., 26)
artistic intent - as music is an art form, it is reasonable artistic intent as an integral element. This can refer to either a composer's intent in assembling sounds as music, or a listener's intent of listening to sounds as music.[1]

The use of this particular system of musical element identification is apt for this creative process, in that soundscape ecology analysis is described through the acoustic properties of sound, as well as cognitive elements related to soundscape recognition and interpretation, which can be related to structure and artistic intent.

Comparing these musical elements then with the parameters of soundscape ecology, a number of relationships might be established:

[1] As Burton argues, "the art of the music composer is to affect the listener by structuring or organisation sound. The art of the (music) listener is to organise sound into meaningful patterns; whether the patterns are intentional, as when listening to a symphony, or environmental, as when listening to the gentle, rhythmic tinkle of wind chimes or John Cage's *4'33"*." (ibid.)

Soundscape Ecology Parameter	Musical Element(s)	Potential Applications
Frequency Patterns	Pitch, timbre	Pitch, timbre, melody, harmony, orchestration
Temporal Patterns	Duration	Pulse, rhythm, duration, tempo, patterns, cycles, structure
Amplitudinal Patterns	Loudness	Dynamics
Spatial Patterns (and gradients)	Spatial location	Acoustics, performance presentation (spatialisation, sound sources, staging)
Interactions	Texture	Counterpoint, orchestration, structure

Table 1: Correlation of soundscape ecology parameter and musical elements.

Consequently, this basic framework establishes associative relationships between soundscape and musical resources, providing a means of identifying the various qualities of a recorded sound or data structure and connecting it with potential musical treatments.

2.3.2. Transformation

There are many techniques that one may employ to develop and transform environmental materials. At a basic level, for acoustic media, these may include transcription of sonic materials or gestures, and sonification/mapping of data; likewise, for electroacoustic media, typical manipulation and editing of the sound samples with various softwares and hardwares occurs, as well as sonification and eco-structural treatments of data. Additionally, novel transformational processes made be developed through adaptation of various environmental dynamics related to the specific place.

2.4. Composition

Once the materials have been developed, the composition of the work may take place.

Here, where the assemblage of the transformed materials takes place, attention can return to the initially sketches and plans, as well as observed ecological processes and relationships, which engender context between the content of the soundscape. These can be used as a guide for composition, and will often relate to the middle to large scale processes or forms of the work (with respect to time scales, communication exchange, and dynamic geographical, climatic, behavioural and morphological processes).

The end result of the composition should be a cohesive work that creatively represents the place, whilst also reflecting upon and responding to the identity of the place.

2.5. Realisation

Once composition is completed, the work is ready for realisation. Preparation and execution of this is with respect to the type of media. For works for acoustic media, the traditional method of delivery to materials to performers, rehearsal(s) and performance(s) remains unchanged, as does the installation, auditioning and presentation of electronic works. These considerations combine in the case of works that make use of both types of media.

2.6. Reflection

The final step of the process is reflection, where evaluation of the process and outcome of the work can determine strengths and weaknesses of the approach and the reception of the final product by an audience. These insights can then inform the way one might approach future compositions.

3. CASE STUDY: MOBILONG SWAMP

3.1. Background

Mobilong (from the Ngaralta clan, of the Ngarrindjeri nation, word *moopolthawong* - 'haven for birds') is part of the Lower Murray Reclaimed Irrigation Area (LMRIA), and is northerly adjacent to the Rural City of Murray Bridge, South Australia. The area was originally a natural floodplain, with the water table level subject to seasonal fluctuation (rise and fall) in the Murray River. This cycle replenished the related aquifers, with the highly saline groundwater subsequently discharging into the river basin.

European settlement saw reclaiming of the floodplains for irrigation, and intent for the Murray to remain a constant water source (for agriculture and navigation) saw the introduction of locks, barrages and levee banks along the river channel. This understandably interrupted the natural cycle, with the groundwater discharge diverted to the irrigation areas themselves. Drainage channels and irrigation cycles artificially replicated the natural process, but multiple droughts in between 2000-2010 saw significant drops in the river level and water table, in turn impacting irrigation activities via reduced water allocations. This in turn allowed the saline groundwater to rise and acidify the heavy clay soil profile, compromising agricultural and farming activities in the area.

The acquisition of much of Mobilong by SA Water has seen the trial of various remedial activities, including revegetation with native eucalypts and plans for diversion of the now acidic discharge water through a nearby local wetlands for natural filtration on route to the river (Sims 2013).

Consequently, the environmental, ecological and sociocultural histories of this area result in its current biodiversity, and the related soundscape that arises through ecosystem function.

3.2. Engagement

As Mobilong Swamp is little more than a kilometre away from my home in Murray Bridge, I have been able to engage with it over an extended period of time. Whilst some of the initial inspiration for writing a work in response to the site came a couple years ago, it was only in the latter half of 2015 that more thorough engagement occurred. This involved a variety of approaches that include community consultation, review of published literature, and on-site observation.

Through contacting with the Rural City of Murray Bridge's environmental officer, I was referred on to a local contractor (previously a dairy farmer at Mobilong) who put me in contact with project managers from SA Water and DEWNR, and researchers from the Waite Institute and Acid Sulphate Soil Centre, University of Adelaide. Through this, I was able to develop an understanding of the ecological evolution of the space, especially in the past decade. Additionally, I spoke with previous landowners (fifth generation farmers), who were able to provide historical perspectives of both European and Indigenous uses of the land.

This contact with the various groups aforementioned provided access to published literature on Mobilong Swamp. These were primarily in the form of departmental reports and independent research on the place's environmental circumstances, but also survey maps care of the local council. Additionally, a list of local birds was sourced from a nearby birdhide, significantly reducing time in identifying species.

The proximity of Mobilong Swamp to my home allowed for ongoing visitation and monitoring of the site quite freely. This consisted of both visits as an individual, which involved observation, documentation and recording of the place, as well as with contractors, environmental scientists and researchers (in their own field work), which allowed for explanation of ecological and environmental circumstances in place.

From this, I was able to develop an understanding of the area's ecosystem and soundscape. Some main points of consideration that emerged include soil conditions, identification and analysis of flora, fauna and habitat, and related sonic spatial distribution.

The salinisation of large soil patches resulting from irrigational land use and recent droughts has directly impacted on the species of flora able to survive in the area. The most impacted areas are now largely populated by samphire and saltbush, by comparison to more alluvial soils which support various grasses and reeds (particularly *Phragmites australis*). As part of ongoing land management, numerous trial plantations of eucalypts (river box (*e. largiflorens*) and river red gum (*e. camaldulensis*) and other natives have been introduced to maintain soil conditions in less impacted areas.

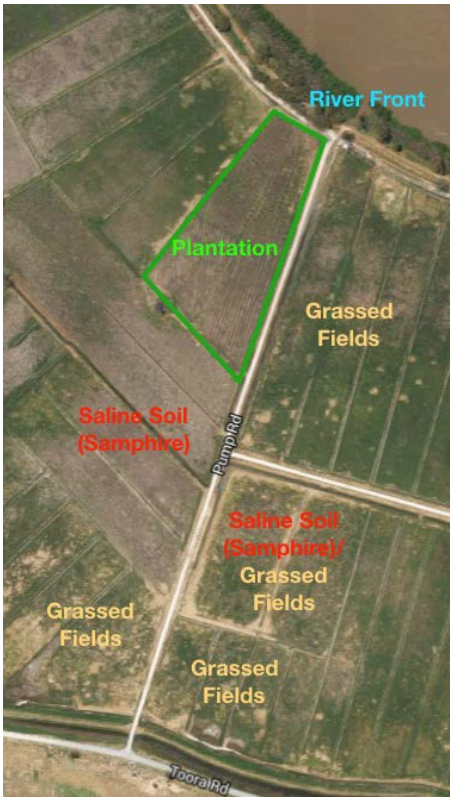


Figure 1: Overview of Pump Road, Mobilong Swamp and adjacent land qualities (Image: Google Maps)

The distribution of flora, combined with boundaries including fences, channels and transport infrastructure, creates distinct habitats with associated fauna. These appear to be in three general types: open field, occupied by small song birds, birds of prey and crickets; plantation, which is populated with a greater variety of small to medium perching and song birds; and river front, which has a notable presence of water birds and frog species.

Each of these overlapping geophonies and biophonies in turn creates distinct sonotopes, through which an observer can pass experience subtle but dynamic changes in the soundscape.

3.3. Creative Response

Essentially derived from the sound walk experience of Pump Road, an 880 metre stretch of unsealed road which bisects the swampland, the overall form of the piece is a idealised procession of the observer (audience) along the road towards the river, in line with the setting sun's shadow cast by an abutting cliff face (at the foot of Pump Road, to the west). Either side of the audience are speakers in front, beside, and behind; between each of these are four miniature string ensembles, creating distinct sonic fields comparable to the fields where much of the soundscape activity occurs either side of the road (Fig. 2). The sonic materials progress from the front of the performance space to the rear (alternating between electronic and acoustic media), evolving over time as new sonotopes within the soundscape are passed through.

FL Speaker		**FR Speaker**
FL String Ens.	**Audience**	FR String Ens.
CL Speaker		**CR Speaker**
RL String Ens.		RR String Ens.
RL Speaker		**RR Speaker**

Figure 2: Performance spatialisation layout, consisting of both electronic and acoustic elements.

There are a number of ecological processes that have been integrated into the responding creative work, including relationship between cricket choruses and temperature, dusk chorus activity, and the impact of habitat alteration on the soundscape.

It has been long known in folklore and scientifically verified that temperature effects the chirp rate of crickets (Walker 1962). At Mobilong, the aforementioned cliff casts a shadow at dusk gradually proceeding towards the river front, creating variation between illuminated and dimmed areas in addition to the cooling of air temperature. As many where many field crickets call at dusk, the resulting choral drone gradually descends, providing a subtle shifting pitch centre.

Additionally, many soundscapes' biophonies are recognised as being most active at dawn and dusk. In addition to crickets, the many species of birds and frogs at Mobilong collectively vocalise as the day draws close to twilight.

This is coupled with the progression through the space's sonotopes. Through change in habitats resulting from Mobilong's restorative management in recent years, the sound walk experience along Pump Road proceeds first through open fields, before passing the plantation and dense reed beds to finally arrive at the rich biodiversity of the river front. Creatively, this experience provides a natural crescendo that is utilised in the form of the responding work, building from a sparse beginning to a climactic finale.

The compositional process began with work the electronic component. Essentially a reconstruction of the soundscape in Ableton Live 9, this utilises recordings made on site in each of the identified sonotopes, with transitions between material determined by temporal calculations relating observer pace to sonotope exposure. Each of the six tracks (front, mid and rear left and right) was further broken down into strata of fore-, mid- and background, allowing for layering of materials based on proximity to the listener.

This was followed by the assemblage of the score for each miniature string ensemble, made up of violins, violas, and cellos. Devised as a modular score where each performer plays the transcribed materials at their own discretion (resonating with the agency of active soundscape participants), the content is comprised of transcriptions of bird song, cricket stridulations and geophonic activity (wind blowing through reeds and trees). Each ensemble's content is, like the electronic counterpart, temporally correlated with sonotope exposure, using time cues to remain in sync. Transcription of the various organisms' calls was done through spectral analysis using the program SPEAR, which permitted a great comprehension of each vocalisation's spectral profile and possible instrumental representations (through the use of standard and extended string techniques).

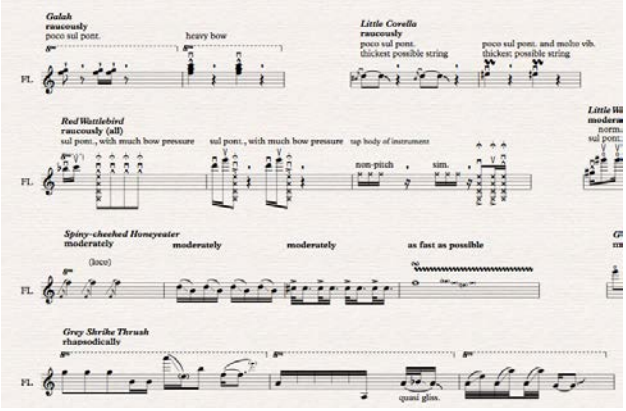


Figure 3: Example from score, featuring various bird calls from which the performer may choose from.

4. REFLECTION

In aspiring to represent soundscapes and ecosystems through ecoacoustic creative work, it is important to recognise that any attempt at exact reproduction is principally flawed. Any creative process that involves collation and development of materials requires multiple decisions that increasingly abstract from the direct soundscape experience, where content (sound and related source) is situated in a environmentally meaningful context (relevant ecosystem and soundscape). This includes not only instrumental resynthesis, which is confined to the timbral and performable limits of the instruments, but also electronic recording and reproduction, which involves many elements (equipment, recording duration and direction, signal processing, to name a few) that subtly distort the original acoustic source and surrounding context through each morphological iteration (O'Callaghan 2012). Performance spaces and circumstances may add further levels of abstraction.

Recognising and adapting one's approach in relation to these issues can in turn provide creative flexibility, whereby one can redirect their attention towards meaningful mimetic or formal representation, rather than exacting authenticity.

To this end, a number of compromises were made for artistic and logistical purposes:

temporal cycles - Like all spaces, the area is subject to meso- and macrolevel temporal cycles, in the form of seasonal and climatic changes and associated dynamics (flooding, solar activity, migration, etc.). In the interest of creating a broad-scale representation of place, rather

than of a specific time period, certain environmental processes were disregarded to preserve creative concept (i.e. the sun's fluctuating azimuth, which for periods of the year does not fall behind the cliff) or heighten performance possibilities (transcribing as many bird vocalisations as possible, disregarding seasonal migration).

pitch materials - a number of vocalisations are pitched beyond the standard range of the violins, and thus a number of these calls are transposed an octave lower to remain playable, with occasional *(poco) sul ponticello* bowing directions intended to provide appropriate higher partials.

call selection - occassionally, calls of certain bird species were deemed too complex to transcribe for the capabilities of the available instrumentation, or too variable/improvisatory to provide meaningful instructions within the established performance methods, and were therefore not included.

anthropophonic sounds - a number of human produced sounds featured in the Mobilong soundscape, including production from a nearby meatworks, passing vehicles on roads, watercraft on the river and overhead aircraft. Additionally, there were various pieces of infrastructure that were thought initially to produce sounds - measuring bores (with wind passing over the opening), powerlines (wind passing through the wires), and an irrigation pump (occassionally operated) - but were either absent/ineffective in the soundscape experience, or too obtrusive in the latter case to warrant inclusion for this work.

From preliminary research and onsite activity to creative analysis and synthesis, and finally realisation of the compositional response, the Mobilong test case study demonstrates the viability and usefulness of adapting soundscape ecology principles to ecoacoustic composition. Moving past superficial engagement with place that might result in impressionistic or highly abstract, mimetically-unrelated responses, and also approaches that centre on specific environmental interests, the discussed framework offers a comprehensive and integrated methods of engaging with place, and related ecosystems and soundscapes.

From this initial investigation , further refinement of the framework will ensue, with additional application through responses to four contrasting South Australian ecosystems (river - Long Island, Murray Bridge; coast - Murray Mouth; sylvan - Adelaide Hills; arid - Farina, Far North SA).

REFERENCES

Allen, A. 2013. "Active listening via soundwalks." *Ecomusicology Newsletter* 2 (2):14-15.

Burton, R. 2015. "The elements of music: what are they, and who cares?" In *ASME XXth National Conference Proceedings*, edited by J. Rosevear and S. Harding. Adelaide.

Farina, A. 2014. *Soundscape ecology: principles, patterns, methods and applications.* Dordrecht: Springer Science+Business.

Farina, A. 2014. "Soundscape ecology: the new frontier in ecology by Almo Farina.". Available online at www.youtube.com/watch?v=DMJ3Fi5SgmQ. Accessed March 2016.

Farina, A., and N. Pieretti. 2012. "The soundscape ecology: A new frontier of landscape research and its application to islands and coastal systems." *Journal of Marine and Island Cultures* 1 (1):21-26.

Klingbeil, M. 2009. *SPEAR: Sinusoidal Partial Editing Analysis and Resynthesis.* Available online at http://www.klingbeil.com/spear/ . Accessed March 2016.

Krause, B. 1993. "The niche hypothesis: a virtual symphony of animal sounds, the origins of musical expression and the health of habitats." *The Soundscape Newsletter* 06 (June).

Krause, B. 2002. *Wild soundscapes: discovering the voices of the natural world.* Berkeley, California: Wilderness Press.

Krause, B. 2015. *Voices of the wild: animal songs, human din, and the call to save a natural soundscape.* New Haven, CT: Yale University Press.

Monacchi, D. 2011. "Recording and representation in eco-acoustic composition." In J. Rudi, ed., *Soundscape In The Arts.* Oslo: NOTAM. pp. 227-250.

Monacchi, D. 2013. "Fragments of Extinction: Acoustic Biodiversity of Primary Rainforest Ecosystems." *Leonardo Music Journal* 23:23-25.

O'Callaghan, J. 2012. "Mediated mimesis: transcription as processing." *Electroacoustic Music Studies Network Conference: Meaning and Meaningfulness in Electroacoustic Music 2012.* Stockholm. Available online at http://www.ems-network.org/IMG/pdf_EMS12_ocallaghan.pdf.

Pijanowski, B., and A. Farina. 2011. "Special edition: soundscape ecology." In *Landscape Ecology* 21 (6): 1201-1344.

Pijanowski, B., L. Villanueva-Rivera, S. Dumyahn, A. Farina, B. Krause, B. Napoletano, S. Gage, and N. Pieretti. 2011. "Soundscape ecology: the science of sound in the landscape." *BioScience* 61 (3):203-216.

Raven Pro: interactive sound analysis software (version 1.5). Ithica, NY: The Cornell Lab of Ornithology. Available online at http://www.birds.cornell.edu/brp/raven/RavenOverview.html.

Schafer, RM. 1993. *The soundscape: our sonic environment and the tuning of the world.* Rochester, Vermont: Destiny Books.

Sims, C. 2013. *Mobilong and Toora Irrigation Areas Condition Assessment Report*. Adelaide: SA Water.

WaveSurfer. Stockholm: Centre for Speech Technology, KHT.

Sueur, J., T Aubin, and C Simonis. 2008. "Seewave: a free modular tool for sound analysis and synthesis." *Bioacoustics* (18):213-226.

Villanueva-Rivera, L., and B. Pijanowski. 2012. "Pumilio: a web-based management system for ecological recordings." *Bulletin of the Ecological Society of America* (93):71-81.

Westerkamp, H. 2002. "Linking soundscape composition and acoustic ecology." *Organised Sound* 7 (1):51-56.